MELIKE DEMIR

Software Engineer

Ankara, Turkey | +90 542 510 59 52 | melikedilekdemir@gmail.com
linkedin.com/in/melike-demir-331827285 | github.com/melogpt | melogpt.github.io



SUMMARY

Versatile and results-driven Software Engineering graduate from Çankaya University with strong hands-on experience in full-lifecycle software development, distributed architectures, and scalable systems. Proven track record across modern tech stacks spanning high-performance web applications (React, TypeScript, Supabase), native mobile solutions (iOS, Swift, SwiftUI), and automated test suites (Python, Pytest, YOLO, OpenCV). Experienced in integrating machine learning pipelines, building asynchronous workflows, and establishing automated QA verification standards, alongside active aerospace engineering research with TUSAŞ. Passionate about clean code, modular software design, CI/CD automation, and delivering robust, user-centric software solutions within agile teams.

EXPERIENCE

Software Engineering Intern | **Mastersoft** **Jul 2025 – Aug 2025**

Hacettepe Teknokent, Ankara, Turkey | *React, TypeScript, ElasticSearch, RESTful APIs, State Management, Git, Agile/Scrum*

- Contributed to the LogAI telemetry platform, developing real-time frontend visualization modules and analytical dashboards for machine learning-driven log anomaly detection pipelines.
- Integrated ElasticSearch distributed query endpoints with asynchronous data fetching patterns, optimizing high-volume search latency and large-scale log aggregation display.
- Architected reusable, type-safe UI component libraries in React and TypeScript, ensuring strict accessibility, responsive state synchronization, and modular code maintainability.
- Participated in bi-weekly Agile/Scrum sprints, contributing to code reviews, API integration testing, and collaborative Git branching workflows.

Software Engineering Intern | **Simtek Simulation Technologies** **Jul 2024 – Aug 2024**

Ankara, Turkey | *C++17, STL, Object-Oriented Design (OOP), Command Pattern, Memory Profiling, Linux, Git*

- Developed core system modules for simulation software tools utilizing modern C++ (OOP, STL, RAII), emphasizing deterministic execution and low-latency state updates.
- Architected and implemented a reversible transaction engine using the Command Design Pattern, enabling multi-level undo/redo operations and state serialization across simulation sessions.
- Engineered grid-based simulation data structures and event-driven logic for internal engineering validation suites, enforcing strict memory profiling and leak prevention.
- Applied collaborative Git workflows, modular architecture principles, and algorithmic optimizations within a professional simulation engineering environment.

FEATURED PROJECTS

Helicopter Display Unit (HDU) Test Automation System **Oct 2025 – Present**

Python, Pytest, Pytest-qt, YOLO (Ultralytics), OpenCV, Tesseract OCR, PyQt5, ReportLab, Speech-to-Text | *TUSAŞ LIFT UP*

- Engineered an automated end-to-end verification platform for Helicopter Cockpit Display Units (HDU) and Warning/Caution/Advisory (WCA) interfaces.
- Trained and deployed YOLO object detection and OpenCV/OCR pipelines to automatically validate graphical avionics indicators, gauge numerical values, and alert color thresholds from screen captures against functional specifications.
- Built an automated test execution suite with Pytest and PyQt5, incorporating automated ReportLab PDF test report generation with visual discrepancy metrics and pass/fail audit logs.
- Implemented speech recognition and audio synthesis modules (SpeechRecognition, pyttsx3), enabling hands-free voice-commanded cockpit test scenario execution.

StayConcierge – Hotel Reservation & QR Housekeeping PMS **2026**

React 19, TypeScript, TanStack Router/Query, Supabase (PostgreSQL, Realtime, RLS), Tailwind CSS, html5-qrcode, Recharts

- Architected a full-stack B2B SaaS platform featuring dual portals: an administrative front-desk PMS dashboard and a mobile-optimized staff web application.
- Engineered a physical QR code room verification workflow using device camera streams (html5-qrcode), allowing housekeeping staff to log room turnarounds with photo proof.
- Configured Supabase backend with PostgreSQL migrations, Row-Level Security (RLS), and Realtime WebSocket listeners, syncing room statuses instantly with front desk.
- Built interactive room occupancy and staff task analytics using Recharts, state management via Zustand, and strict form schemas with Zod.

FEATURED PROJECTS (CONTINUED)

SkinCare AI – Personalized Dermatological Routine & Scanner 2026

Swift, SwiftUI, iOS 18+, Vision AI, AVFoundation, Prompt Engineering, OpenAI/Gemini API, Swift Concurrency, MVVM

- Architected a production-ready iOS 18+ personalized skincare assistant with SwiftUI and MVVM, utilizing Swift Concurrency (async/await) for background analysis.
- Engineered an on-device OCR and product label scanner using Apple Vision and AVFoundation to extract cosmetic ingredient formulations in real time.
- Designed a multi-layered Derma Blueprint prompt engineering architecture that injects user skin profiles (dry, oily, sensitive) into LLM context to evaluate chemical conflicts (e.g., Retinol vs. AHA/BHA) and generate safety assessments.
- Constructed an FDA/EMA-aligned medical disclaimer onboarding gate alongside a zero-latency native iOS launch screen and interactive routine tracking.

BirdLens – AI Bird Identification & Diagnosis 2026

Swift, SwiftUI, StoreKit 2, OpenAI API, Vision / CoreML, MVVM, App Store Sandbox & Production

- Built a commercial-grade iOS application for instant bird species identification and multimodal symptom-based avian disease diagnosis.
- Integrated native Apple StoreKit 2 in-app purchase infrastructure supporting Weekly, Annual, and Lifetime subscriptions with custom paywall comparison UI, transaction verification, and freemium quota limits.
- Implemented diagnostic workflows supporting combined image and text symptom queries, incorporating on-device classification fallbacks for offline reliability.

TECHNICAL SKILLS

Languages:	Swift, Python, TypeScript, JavaScript, C++, C, Java, SQL, HTML5, CSS3
Mobile Development:	SwiftUI, iOS 18+ SDK, StoreKit 2 (Subscriptions & IAP), MVVM, Swift Concurrency (async/await), Vision Framework, AVFoundation, WatchKit, UserNotifications
Web & Full-Stack:	React 19, TanStack (Router, Query, Start), Supabase (PostgreSQL, Realtime, RLS), Node.js, Tailwind CSS, REST APIs, Zustand, Zod, Vite
AI & Computer Vision:	YOLO (Ultralytics), OpenCV, Tesseract OCR, LLM Prompt Engineering (OpenAI, Gemini), Multimodal AI, CoreML
Testing & Automation:	Pytest, Pytest-qt, PyQt5, ReportLab (PDF Test Reports), Git, GitHub CI/CD, Xcode, Cursor IDE
Analysis & Methodology:	Agile & Scrum, Process Modeling (UML, Flowcharts, BPMN), SRS / SDD Preparation, Requirements Gathering & Documentation, WBS, User Stories

EDUCATION

Çankaya University, B.Sc. Software Engineering 2021 – 2026

Ankara, Turkey

Comprehensive curriculum covering Software Architecture, Algorithms, Object-Oriented Systems, and Intelligent Software Engineering. Capstone graduation project developed in formal partnership with TUSAŞ LIFT UP defense research program.

CERTIFICATIONS

- **19th R&D Project Market (19. Ar-Ge Proje Pazarı):** Certificate of Acceptance & Exhibition for TUSAŞ LIFT UP Helicopter Display Unit Test Automation Project.
- **Geleceği Kodluyoruz:** IEEE Computer Society / LST Software Development Track.
- **Meta:** Introduction to Front-End Development Specialization.
- **Turkcell Geleceği Yazanlar:** Swift ile iOS Programlama Başarı Sertifikası.

COMMUNITY & COMPETITIONS

- **Apple WWDC 2026 Attendee:** Participated in worldwide developer sessions covering Swift Concurrency, SwiftUI enhancements, StoreKit 2 architecture, and iOS 18 SDK capabilities.
- **TUSAŞ LIFT UP Industry Mentorship:** Selected participant in aerospace industry-academia R&D program, collaborating with flight test and software verification engineers.
- **Cursor Hackathon Turkey (Oct 2025):** Architected and demonstrated an AI-driven language learning mobile application prototype within an intensive agile hackathon setting.
- **University Tech Communities:** Active contributing member of IEEE Computer Society and Google Developer Student Clubs (GDSC) at Çankaya University.

LANGUAGES

Turkish (Native) | English (Professional Working Proficiency) | German (Beginner)